
Software Test Plan Document

System Name : SeedOfLife :
Mobile Application for Plants Care.

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DOCUMENT INFORMATION AND APPROPOVALS

VERSION HISTORY			
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1.0	/3/2025	Mohamad Farris Foo Amirul Ehsan	Draft Software Test Plan
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DOCUMENT APPROVALS			
Approver Name	Project Role	Signature/Electronic Approval	Date
Siti Fatimah Omar	Lecturer		

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i. Purpose of Software Test Plan Document

The purpose of this document is to guarantee that the software meets the client's expectations based on the specified, well-defined requirements. Additionally, this document also aims to validate the flexibility, efficiency, and robustness of the system as well as to ensure proper adherence to the standard compliance and regulations. This Software Test Plan (STP) will also serve as a planning framework and guideline for testing prior to implementation.

ii. Contents of Software Test Plan Document

A test plan contains the execution strategy for a testing session. It provides a detailed specification of planning in terms of many aspects such as testing policy, principles, mission, objectives, scope, risks, strategy, design technique, participating team, test environment, resources, entry and exit criteria, risk assessment and analysis, as well as test prioritization. This STP also provides explanations on each module and its functionalities.

iii. Topics to be Covered in Software Test Plan Document

The items that are proposed in this document should cover the test which describes: the testing approach, scope, risk and contingency, entry and exit criteria, item pass/fail criteria and many more. The Test Specification highlights and describes the current test cases in development in simpler forms and includes use case diagrams, use case statements and test case specifications. Additionally, the Test Environment describes all the required testing sites, hardware, and software for the tests to be executed smoothly.

To satisfy all project requirements, every item developed in this document will address the items that are outlined and specified in the SRS (Software Requirement Specification).

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1. TEST POLICY

This section explains the major philosophy of the organization of Vitamin Lab (dummy organization) regarding testing and its terms, meaning and practices. However, most of these policy aspects are widely applied in any other organization even UniKL MIIT may have this in their test policy since this is based on ISTQB (worldwide standard).

1.1 Software Testing Term Definition

Testing is a development activity that is all about the process of finding and identifying defects by performing a test and checking if the actual result matches the expected results or not. It also known as process of verification and validation which is based on ISO-9000 standard as shown below:

- **Validation** - Confirmation by examination and through the provision of objective evidence that specified requirements have been fulfilled. To summarize ISO statement, validation confirms that product build with correct requirement that satisfy the customer requirement.
- **Verification** - Confirmation by examination and through provision of objective evidence that the requirements for a specific intended use or application have been fulfilled. To summarize the ISO statement, verification confirms that software has been created and fulfilled.

Testing also defines what will happen once a defect occurs (failure). Testing is performed by the testing team which then the report/result will be forwarded to the developer team so that they can start debugging.

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1.2 Advantages of Testing

There are several significant advantages of testing that can produce magnificent software application with supreme quality. These are numerous advantage that can be extract from testing:

- Testing increases quality in terms of the software attribute indirectly in which defect can be detected and corrected earlier prior to delivery. In addition to that, testing did increase process quality as well since defects are documented which means it can be analyzed, processed and solution to that error can be reused again to prevent same error again.
- Testing boosts confidence of all stakeholders which affects perception of team management regarding the software quality as well its robustness due to very few or no failure are found.
- Testing is like liability insurance where it costs a lot of money, but it ensures all the stakeholders to be at ease as it covers and mitigates all the possible risks.
- Testing that is successfully conducted leads to major reductions in cost as defects can cause huge costs.

1.3 Testing Principles

There are seven testing principles that this organization applies which are based on ISTQB standard.

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Principle 1: Testing shows presence of defects.

Testing can determine and reveal defects are present but unable to verify and identify that there are no defects at all are presents. According to Edger W. Dijkstra, 1970 “Program testing can be used to show presence of bugs, but never to show their absence!”

Principle 2: Exhaustive testing is impossible.

Full complete/exhaustive testing is truly not feasible except for trivial situations. Risk-based testing will be used to produce risk analysis to determine priorities of features module that need to be carried out.

Principle 3: Early testing

Testing is not a phase that happens at the end of software development, it should start as early as possible. Early testing (like static testing; review) takes place parallel to development activities, defects are detected earlier thus cost less to fix.

Principle 4: Defect clustering

Testing effort shall be focused proportionally to the expected and observed defect density of modules. A small number of modules often contain most of the defects discovered during pre-release testing/or responsible for most of operational failures.

Principle 5: Repetitions have no effects – Pesticide Paradox

Repeating tests bring no new insight thus test cases need to be regularly reviewed, revised, and modified. So that we won't execute the same test and test cases repeatedly, there is a need for changes.

Principle 6: Testing is context dependent.

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Testing depends on the context usage of that software. For example, safety-critical software systems are TESTED DIFFERENTLY (more intensive in technique) from commercial applications.

Principle 7: Absence of errors fallacy

System without failures does not imply system will meet user expectations. Sometimes there are no errors and failures, but the functionalities given are not what the user wants and expected.

1.4 Test Missions & Objectives

The SeedOfLife team is committed to delivering a high-quality, reliable, and user-friendly mobile application by implementing a rigorous testing strategy. Our testing mission ensures that functional and non-functional requirements are met, defects are minimized, and the app performs optimally under real-world conditions in Malaysia.

1.5 Communication Among Stakeholders

Effective stakeholder communication is critical to ensure SeedOfLife meets quality standards, aligns with user needs, and resolves issues efficiently. The testing process involves collaboration among developers, testers, product managers, designers, business analysts, and end-users—all contributing unique insights to refine the app.

1.6 SDLC Identification

“Seed of Life” adopts the Prototyping Model as its Software Development Life Cycle (SDLC). This model emphasizes iterative development through the creation and refinement of preliminary prototypes based on continuous user feedback. By focusing on early visualization and testing of core functionalities, the approach ensures that usability challenges are identified and addressed quickly while evolving features like AI-based plant identification are seamlessly integrated. The Prototyping Model is particularly effective for SeedOfLife as a user-centric plant care app, enabling rapid

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adaptation to urban gardeners’ needs while addressing real-world challenges in Malaysian high-density living environments.

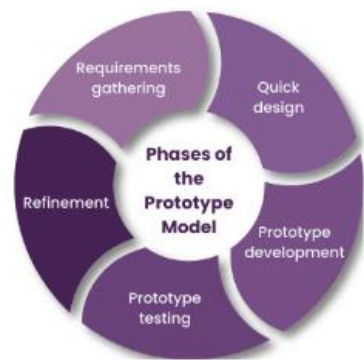


Figure 1: Prototype Model Diagram

Phases	Description
Requirements Gathering	This phase involves collecting functional and non-functional requirements from target users. For the SeedOfLife application, this includes plant care preferences, localized climate considerations, and usability needs specific to urban gardeners in Malaysia.
Prototype Development	Low-fidelity prototypes, such as wireframes and mockups, are developed using tools like Figma. These prototypes simulate key functionalities like plant identification, care scheduling, and nursery locator features.
User Evaluation	Prototypes are shared with users to gather feedback on usability, functionality, and design. This feedback is used to iteratively refine the prototype before moving to full development.

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Implementation	Once validated, the prototype evolves into the final application using tools such as Flutter for the frontend and Firebase for backend integration.
Testing and Maintenance	Comprehensive testing—including unit, integration, and beta testing—is performed to ensure smooth functionality. Ongoing maintenance incorporates user feedback for continuous improvement.

Table 1: Prototype Model Description

By employing the Prototyping Model, the SeedOfLife application ensures flexibility, adaptability, and user satisfaction throughout the development lifecycle. This methodology is well-suited to the project’s objectives of creating a robust and user-friendly plant care solution for urban Malaysian gardeners.

2. INTRODUCTION TO TEST PLAN

This document serves as the official Acceptance Test Plan (ATP) for the SeedOfLife mobile application. The ATP provides a structured framework for the quality assurance team to validate that all functional and non-functional requirements have been properly implemented according to the Software Requirements Specification (SRS).

The acceptance testing will employ black-box testing methodologies, focusing on verifying system behavior against documented requirements without examining internal code structures. Test cases will be derived directly from use case specifications contained in the SRS, ensuring comprehensive coverage of all key user scenarios. This includes but is not limited to: plant identification accuracy, care schedule generation, localized weather adaptation, and community feature functionality.

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The testing process will reference multiple project documents including the SRS, use case diagrams, UI/UX design specifications, and technical architecture documents. All test artifacts including test cases, scripts, and results will be properly documented and version-controlled to maintain traceability throughout the testing lifecycle.

This ATP establishes the foundation for formal acceptance testing that will validate SeedOfLife's readiness for production deployment and ensure it meets all stakeholder requirements for functionality, usability, and performance in real-world Malaysian urban gardening scenarios.

2.1 System Description

SeedOfLife is a mobile application designed to simplify plant care for urban residents in Malaysia, especially those living in apartments. The app uses AI-powered image recognition to identify tropical plants commonly found in Malaysia and provides personalized care instructions tailored to local climate conditions. Key Functionalities Include:

Key Functionalities Include:

- Plant Identification:
 - Capture/upload plant photos for AI-based species recognition
 - Access detailed plant profiles with care instructions
- Personalized Care Management:
 - Customized watering, fertilizing, and pruning schedules
- Community Platform:
 - Forum for sharing tips, asking questions, and troubleshooting
 - Moderated discussions categorized by plant type and care topics

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- Nursery & Shop Locator:
 - GPS-based map of nearby nurseries and gardening stores
 - Filter by store type, product availability, and user ratings

- Plant Database:
 - Searchable library of 500+ plants suitable for Malaysian homes
 - Difficulty levels, light requirements, and common issues

User Roles and Capabilities:

- Guest Users:
 - Basic plant identification
 - Access to general care information
 - Option to register for full features

- Registered Users:
 - Save plants to “My Garden”
 - Set up personalized care schedules
 - Participate in community forum
 - Use nursery locator with filters

- Administrators:
 - Moderate forum content
 - Verify nursery listings
 - Update plant database entries

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2.2 Assumptions

To conduct and execute the test, it assumed that:

Ref	Assumption
1.	Functional requirements have been reviewed and acknowledged.
2.	There is a Software Requirements Specification (SRS) as reference where it contains all the functional requirements.
3.	Static test has been conducted via observation, reading and review of tested document (SRS)
4.	The test resources and test environment described in section 6 is available
5.	System, database, and server able to run at smooth performance
6.	Identified defects and bugs are expected to be fixed/resolved before the testing
7.	All changes to requirements will be notified and communicated to test team

Table 2: Assumptions

2.3 References

SRS (Software Requirements Specification)	There is a Software Requirements Specification (SRS) to be referenced where it contains all the functional requirements.
IEEE Standard 829 - 1983	IEEE Standard for Software Test Plan.

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Discovery Proposal	A proposal that defines Discovery overall during early planning.
ISTQB Slides	Numerous slides from chapter 1 until 7 are used as references. Obtained from Software Testing class.
ISO/IEC 25010:2011	Reference to define quality requirements and quality attributes.

Table 3: References

2.4 Document Overview

The STP document consists of six sections as brief below:

- Section 1 contains Test Policy components which explain the major philosophy of the development organization regarding their testing and its terms, testing SOP (Standard of Procedure), meaning and practices.
- Section 2 explains Introduction to Test Plan which gives an overview about the current acceptance test plan purpose, system description, references, and document structure overview.
- Section 3 contains Test Strategy which describes the testing approach, test objective, test scope, test process, test level, risk and contingency, entry and exit criteria, item pass/fail criteria, suspension criteria and resumption requirements.
- Section 4 contains Test Case Specification which describes the current test cases in development along with Use Case Diagram.
- Section 5 contains Test Environment which describes all the required facilities, hardware, software, and other necessary requirements for test to be executed smoothly.

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- Section 6 contains Test Schedule, which discusses an overall schedule plan including testing activities.

3. TEST STRATEGY

A test strategy is an outline that describes the testing approach of the software development cycle where it describes how the testing activity will be done. It is the main core part of this test plan.

3.1 Test Objective

The primary objective of this test plan for SeedOfLife is to ensure the mobile application functions smoothly, reliably, and delivers all intended plant care features without issues. Testing will verify that the application meets all requirements specified in the Software Requirements Specification (SRS) while fulfilling stakeholder expectations for an effective urban plant care solution in Malaysia's unique climate conditions.

Here are the main objectives:

1. Verify Plant Identification Accuracy

Test the AI-powered plant identification system to ensure accurate recognition of tropical plant species common in Malaysia.

2. Validate Personalized Care Scheduling

Ensure the system generates and adapts care schedules based on plant type, user location, and Malaysian climate conditions.

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3. Test Community Platform Functionality

Verify users can create posts, comment, react, and report content in the moderated community forum.

4. Validate Nursery Locator Accuracy

Test GPS-based nursery finding with filters for store type, ratings, and product availability.

5. Ensure User-Friendly Design

Confirm the application is intuitive, visually appealing, and accessible to users with varying technical skills.

6. Verify Data Security and Privacy

Perform security checks to ensure user data, plant collections, and personal information are protected.

7. Test System Performance

Check application responsiveness, resource usage, and stability under typical usage conditions.

8. Validate Notification System

Ensure care reminders and community notifications are delivered timely and accurately.

9. Confirm Offline Functionality

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Test core features (viewing saved plants, care schedules) without internet connectivity.

10. Gather Stakeholder Approval

Collect feedback from supervisors and users to confirm the system meets requirements before deployment.

3.2 Scope

The scope of this Acceptance test plan is to ensure that all the functional requirements defined in SRS are met and tested accurately.

3.3 Scope

The item to be tested is Seed of life Application Prototype.

3.3.1 Features Tested

There is a risk-based testing which includes a risk analysis conducted under the risk-based testing section of Risk Analysis 3.9.2.1. The result of that risk analysis determined the features to be tested where it is based on the likelihood, impact, total risk, and priority. Any feature with high attribute of those mentioned earlier will be tested first.

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3.4 Test Process

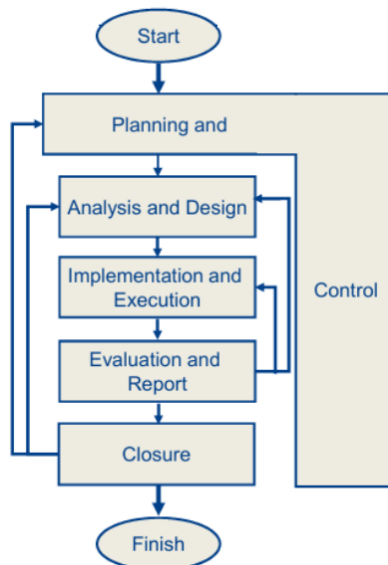


Figure 2 Test Process

Initially, there are five main processes involved in this approach.

1. Planning and Control

❖ Planning:

- Define test objectives, test scope, and testing strategy to align with project requirements.
- Review documents like the Software Requirement Specification (SRS) to identify testable requirements and potential test cases.
- Develop a comprehensive test plan document detailing:
 - Test objectives
 - Test scope
 - Test strategy and schedule
 - Risks and mitigation plans
 - Resources required (personnel, tools, hardware, software)
- Construct test cases based on use cases identified in the SRS

❖ Control:

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- Monitor progress, ensuring the planned testing activities align with the schedule.
- Adjust test plans dynamically based on project updates or identified risks.
- Ensure quality and efficiency by reviewing and refining test cases and procedures.

2. Analysis and Design

❖ Analysis:

- Evaluate the system and its requirements for testability.
- Review the identified test cases and ensure they comprehensively cover functional and non-functional requirements

❖ Design:

- Prepare the testing environment, ensuring tools, software, and hardware are available and functional.
- Arrange entry criteria and ensure all dependencies (e.g., modules, data) are prepared.
- Finalise test cases, scripts, and test data, ensuring clarity and traceability to the requirements

3. Implementation and Execution

❖ Implementation:

- Develop detailed test scripts or procedures based on the finalised test cases.
- Integrate test scripts with tools and infrastructure for seamless execution.
- Verify the readiness of the test environment.

❖ Execution:

- Run the tests according to the defined plan and scripts.
- Compare actual results with expected outcomes to determine pass or fail status.
- Record actions taken, inputs, outputs, and any incidents encountered for documentation purposes.

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4. Evaluation and Report

❖ Evaluation:

- Analyse all identified incidents and defects, prioritising them based on severity and impact.
- Reassess test results against expectations to identify gaps or issues.

❖ Report:

- Share incident and defect reports with relevant stakeholders.
- Ensure issues are addressed, retesting where necessary to verify fixes.
- Document all results, lessons learned, and any challenges faced.

5. Test Closure

- Evaluate the exit criteria to determine if testing objectives have been met.
- Ensure and check deliverables delivered or not
- Closing all incident reports, ensuring proper documentation for future reference.
- Archive Testware, environment and infrastructure for next reuse usage
- Evaluate how testing went and learn lessons for future reference

3.5 Item Pass/Fail Criteria

The test oracle to determine pass and fail in test level is the functional requirements or use case that are based on Seed of Life SRS that consists of requirements and use case specification.

If the test fails to meet the objectives of any of the functional requirements or use cases, then it is considered a failure thus a defect or incident will be documented and reported where it will be reviewed by the tester and developer.

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3.6 Suspension Criteria and Resumption Requirements

3.6.1 Suspension Criteria

Testing will be suspended if:

- The SeedOfLife application crashes consistently during test execution
- Critical defects prevent core functionality testing (e.g., plant identification fails entirely)
- Test environment becomes unavailable (server downtime, database failure)
- Major requirement changes are introduced mid-testing

3.6.2 Suspension Criteria

Testing may resume when:

- Critical defects are fixed and verified
- Test environment is restored and stable
- Updated test cases are prepared for requirement changes

3.7 Entry and Exit Criteria

This section will specify when to start the testing activities and when to stop the testing activities. This is essential and required to be mentioned in the test plan as exhaustive testing is impossible.

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3.7.1 Entrance Criteria

Below are the criteria required to start the testing activities (Execution)

- All required tools, whether in the form of hardware or software for the test environment, are available.
- The test environment has been set up and prepared with reliable quality beforehand.
- Requirements are documented in SRS, reviewed, and approved by relevant stakeholders (supervisor specifically).
- Previous tests in system, integration and unit test level have been conducted and satisfactory.

3.7.2 Exit Criteria

Testing activities may conclude when:

- All high-priority test cases (Priority 4–5 from Risk Analysis) have been executed
- Minimum 80% test coverage achieved across all functional modules
- All critical and major defects have been resolved and retested
- Test summary report is complete with documented results
- Stakeholder approval is obtained for production readiness

3.8 Risks and Contingencies

Several risks related issues need to be addressed with contingency during testing as it could lead to major troublesome problems that will affect testing development. There are project and product risks, for product it will be specified in risk-based testing section while here are the project risks:

ID No	Project Risks	Impact	Contingencies
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1	Plant identification API failure	Unable to identify plants, user frustration	Use cached results, switch to offline mode, notify users
2	Unavailability of test devices	Delay in mobile testing	Use emulators, prioritize critical tests
3	Lack of plant dataset for testing	Inaccurate identification validation	Use synthetic images, partner with botanical gardens
4	Time constraints	Incomplete test coverage	Prioritize risk-based testing, focus on critical features
5	Weather API unavailability	Care reminders not weather-adaptive	Use default tropical climate profiles
6	Community forum spam/content issues	Poor user experience, moderation overhead	Implement auto-moderation, increase moderator support
7	Changes of original requirements	Delay the test	Increase more personnel or effort.

Table 4: Project Risks

3.9 Test Level

The Seed of Life incorporates various testing levels to ensure that each functionality meets its intended purpose. The primary test level to be conducted for this project is Acceptance Testing.

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Acceptance Testing

At this level, the system is evaluated to confirm whether it meets all specified requirements and operates as intended for end-users. This testing phase ensures the application is ready for real-world deployment. To achieve this, a black-box testing approach is utilized, focusing solely on system functionality without examining the internal code. The testing will rely on the Software Requirements Specification (SRS) document as its foundation. It will involve students, coaches, and admins, with the assistance of developers, to validate the viability of the system at the testing site.

3.9.1 Black Box Testing

The black-box testing approach focuses on validating the system's input and output. This method is particularly suitable as it only assesses functionalities without delving into the underlying code. For this project, it is referred to as Specification-Based Testing because it strictly adheres to the SRS document.

3.9.1.1 Functional Testing (Test Type)

Functional testing is performed to confirm that every feature of the system works as intended. Each functionality is tested against its defined behavior in the SRS. This ensures that modules like plant recognition scan, Notifications, and Plant-care Management perform correctly and consistently.

3.9.1.2 Use Case Based Testing

Use Case-Based Testing is an essential method for validating whether SeedOfLife's plant care workflows function correctly in real-world urban gardening scenarios. This approach tests specific interactions between users and the mobile application to achieve key objectives like plant identification, care scheduling, and local resource finding. By converting use cases from the Software Requirements Specification into detailed test scenarios, we can thoroughly evaluate both successful outcomes and potential failure cases. For instance,

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when testing the plant identification feature, valid scenarios would confirm accurate detection of common Malaysian plants like orchids or ferns with proper care instructions, while invalid cases would verify how the system handles poor quality images or non-plant photos. Similarly, testing personalized care reminders would validate both ideal conditions where schedules adapt perfectly to Malaysia's climate and edge cases like unexpected weather changes. This method provides realistic validation of SeedOfLife's core functionality while ensuring all features meet urban gardeners' needs. By examining preconditions (like user login status), actions (such as scanning a plant), and postconditions (like receiving care tips), we can identify any gaps in user workflows. Use Case-Based Testing is particularly valuable for SeedOfLife as it verifies the app's specialized features work harmoniously in Malaysia's unique tropical environment while maintaining an intuitive user experience. This comprehensive approach helps detect issues early, from data inaccuracies in plant databases to flaws in the nursery locator's geolocation functionality, ultimately ensuring the app delivers reliable, localized plant care assistance.

3.9.2 Risk Based Testing

A risk-based testing approach will be established to help prioritise and allocate testing resources. Due to time restrictions and limited resources, this strategy prioritises testing essential sections of the system first.

This testing involves assessing the risk based on likelihood and impact of failures. These are the steps to conduct it include:

- Analyze the requirements and its document.
- Determine risk by calculating the likelihood and impact of each requirement through cost.
- Prioritize risks through the total risk which is likelihood to occur multiply by impact.

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3.9.2.1. Risk Analysis

Risk-Based Testing prioritises functional testing based on the risk of failure and its effect. This technique guarantees that essential system components, such as attendance monitoring and payment processing, are properly tested to prevent catastrophic interruptions. Testing is focused towards locations with the greatest risk ratings, as established in the Risk Analysis Table.

Formula (based on MSTB/ISTQB)

- Likelihood from 1 (negligible) until 9 (very high)
- Impact (non-linear scale of money values [e.g. 1,10,50,100,500,1000])
- Total Risk = $L * I$

Total Risk (r)	Priority Number
$r < 3000$	1 (lowest)
$r \geq 3000$	2 (low)
$r \geq 4000$	3 (medium)
$r \geq 8000$	4 (high)
$r \geq 9000$	5 (highest)

Table 5: Risk Classification for Prioritization

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SeedofLife	Likelihood	Impact	Total Risk	Priority
Features	Part of its reason can be usage frequency, as the more you use, the more bugs discovered	Based on monetary value which determined by its importance	$R = L * I$	Highest Priority: 5 Lowest Priority: 1
Failed Plant Identification (Plant Identification)	8	800	6,400	4
Misdiagnoses common plant diseases (Plant Identification)	7	800	5,600	4
Fails to recognize rare endemic species (Plant Identification)	8	900	7,200	5
Misclassifies similar-looking plants (Plant Identification)	7	800	5,600	4
Struggles with diseased leaf detection (Plant Identification)	6	750	4,500	3
Incorrect Care Recommendations (Smart Care System)	7	1,000	7,000	5
Overwatering during monsoon season (Smart Care System)	6	750	4,500	3
Suggests wrong light for balcony plants (Smart Care System)	5	600	3,000	2
Fails to adjust for air-conditioned environments (Smart Care System)	5	600	3,000	2
Failed Weather-based Reminders (Care Scheduling)	6	700	4,200	3

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Faulty sunlight recommendations (Care Scheduling)	5	600	3,000	2
Nursery Locator Malfunction (Nursery Locator)	5	600	3,000	2
Lists unavailable plants at nurseries (Nursery Locator)	7	650	4,550	3
Poor GPS Accuracy in dense urban areas (Nursery Locator)	6	600	3,600	3
Doesn't show specialty nurseries (Nursery Locator)	5	550	2,750	2
Community Forum Downtime (Community Platform)	4	400	1,600	1
Fails to detect harmful content (Community Platform)	5	500	2,500	2
Poor GPS accuracy for nursery locator	9	1,200	10,800	5
User Profile Data Loss (User Management)	3	900	2,700	2
Login delays during peak hours (User Management)	7	600	4,200	3
Community forum downtime	6	800	4,800	3
AI misidentification of tropical plants	7	500	3,500	3

Table 6 : Risk Analysis for Risk Based Training

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4. TEST CASES

Test cases contain a step-by-step flow of action where it will be performed by testers. Each test case contains tested by whom, test level, test type, test design technique, risk number, test case number, test case name, requirement traceability, test case description, preconditions, post conditions, item(s) to be tested, test procedure specification, input step(s), test data, expected result, actual result, pass/fail, and stakeholder (user).

4.1 Use Case Seed Of Life – Overall Mobile Application

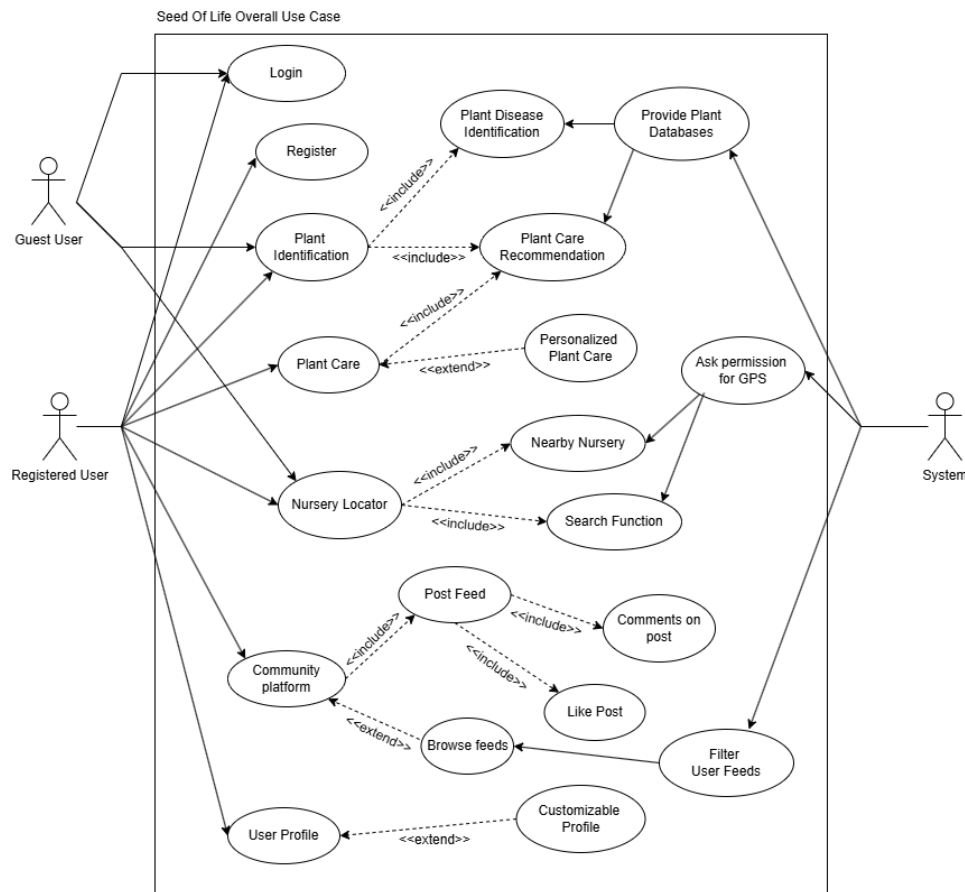


Figure 3 : Use Case Diagram SeedOfLife – Overall Application

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4.1.1 Use Case: Plant Identification Module

Use Case ID	UC_001	
Use Case Name	Capture/Upload Plant Photo	
Description	Allows users to take a photo of a plant or upload an existing image from their gallery for identification.	
Primary Actor	User (Guest/Registered)	
Include use cases	UC_002 (Identify Plant Species)	
Scenario	Step	Action
Main Flow	Pre-Condition	User opens the SeedOfLife app and selects "Identify Plant"
	1.1	User chooses between "Take Photo" or "Upload from Gallery."
	1.2	If "Take Photo," the app accesses the device camera; user captures a clear image of the plant.
	1.3	If "Upload from Gallery," user selects an image from their device.
	1.4	App validates image (e.g., file format, minimum resolution).
	1.5	Image is sent to the AI identification module (UC_002).
	Post-Condition	Plant details are displayed (or top 3 matches if uncertain).
Alternate Flow	Pre-Condition	User attempts to upload an invalid image (e.g., blurry, non-plant).
	2.1	App detects low image quality or irrelevant content.
	2.2	Displays error: "Please upload a clearer photo of the plant."
	2.3	Returns to Step 1 in Main Flow.

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Robust Flow	Post-Condition	User retries with a valid image.
	Pre-Condition	User loses internet connection during upload.
	3.1	App caches the image locally and notifies: "Connection lost. Retry when online."
	3.2	Automatically resumes upload when connectivity is restored.
	Post-Condition	Plant identification proceeds after reconnection.

Table 7: Use Case Statement for Capture/Upload Plant Photo

Test Case for UC_001: Capture/Upload Plant Photo

Tested By	Mohamad Farris Foo	Test level	System Testing
Test Type	Black Box Testing – Functional & Security Testing		
Test Design Technique	Use Case Testing		
Risk Number	2 (Medium)		
Test Case Number	TC_001.1		
Test Case Name	Plant Photo Validation Test		
Requirement Traceability	REQ_F101		
Test Case Description	To verify the system properly handles plant photo capture/upload and validates image quality before identification.		
Item(s) to be tested			
1	Capture/Upload Plant Photo Use Case		
Pre-condition	- User has launched the SeedOfLife app - "Identify Plant" feature is accessible		
Test Procedure Specification			
Input	Test Data	Expected Output/Result	
User selects "Take Photo" and captures a clear	Test image: Monstera deliciosa leaf (well-lit, focused)	System accepts image, displays preview, proceeds to UC_002	

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plant image (500x500px minimum)		
User uploads out-of-focus plant photo	Test image: Blurry snake plant (200x200px)	System rejects image with toast message: "Please upload a clearer photo of the plant."
User uploads a food photo	Test image: Plate of nasi lemak	System detects non-plant content, shows: "This doesn't appear to be a plant. Try again."
Post-condition	- Successful cases: Image proceeds to UC_002 (Plant Identification) - System logs all validation attempts for analytics	

Table 8 : Test Case Statement for Plant Photo Validation Test

Use Case ID	UC_002	
Use Case Name	Identify Plant Species	
Description	The system analyzes the uploaded plant photo using AI to identify the species and returns localized care instructions.	
Primary Actor	User (Guest/Registered)	
Include use cases	UC_001 (Capture/Upload Plant Photo)	
Scenario	Step	Action
Main Flow	Pre-Condition	User has uploaded a valid plant photo (from UC_001).
	1.1	System sends the image to PlantNet/Plant.id APIs for analysis.
	1.2	AI cross-references the image with tropical plant databases (1,000+ Malaysian species).

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	1.3	If confidence $\geq 80\%$, system displays: - Plant name (common + scientific) - Care tips (water, sunlight, soil) - Local climate adaptations (e.g., "Reduce watering during monsoon").
	1.4	User can save the plant to "My Garden" (registered users only).
	Post-Condition	Plant details are displayed; care schedule is generated (UC_004).
Alternate Flow	Pre-Condition	AI confidence is 50-79% (uncertain match).
	2.1	System suggests top 3 possible matches with similarities.
	2.2	Prompts user to: - Upload additional photos (e.g., leaf close-up) - Answer questions (e.g., "Does it have flowers?").
	2.3	Refines identification based on new inputs.
	Post-Condition	Best match is confirmed or user exits.
Robust Flow	Pre-Condition	Plant identification API fails (timeout/server error).
	3.1	System switches to offline mode: - Uses lightweight cached model (200 most common plants). - Notifies: "Limited offline results; reconnect for full accuracy."
	3.2	If no match found, suggests posting to the community forum (UC3.1) for human help.
	Post-Condition	User receives partial results or redirected to community.

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Table 9: Use Case Statement - Identify Plant Species

Test Case for UC_002: Identify Plant Species

Tested By	Amirul Ehsan	Test level	System Testing
Test Type	Black Box Testing – Functional & AI Validation		
Test Design Technique	Use Case Testing + Boundary Value Analysis		
Risk Number	3 (High)		
Test Case Number	TC_002.1		
Test Case Name	Plant Identification Accuracy Test		
Requirement Traceability	REQ_F201, REQ_F202		
Test Case Description	To verify the system accurately identifies tropical plant species and handles edge cases.		
Item(s) to be tested			
1	Plant Identification Module		
Pre-condition	Valid plant image uploaded via UC_001		
Test Procedure Specification			
Input	Test Data	Expected Output/Result	
Clear image of Monstera deliciosa leaf (front/back)	Test image: HQ Monstera with fenestrations	Returns: - "Monstera deliciosa" - "Water: 1x/week in KL condos" Confidence: ≥85%	
Partial image of orchid flower (no leaves)	Test image: Purple Phalaenopsis bloom	Shows: - Top 3 orchid varieties - Prompt: "Upload leaf photo for better ID"	
Simulate API timeout (500ms delay)	Test image: Snake plant	Uses offline model, shows: - "Sansevieria trifasciata (Offline)" - Disclaimer about limited accuracy	
Post-condition	Plant details saved to session storage		

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Table 10: Test Case Statement - Identify Plant Species

Use Case ID	UC_003	
Use Case Name	View Plant Details	
Description	Displays comprehensive plant information and care instructions after successful identification.	
Primary Actor	User (Guest/Registered)	
Include use cases	UC_002 (Identify Plant Species)	
Scenario	Step	Action
Main Flow	Pre-Condition	Plant has been successfully identified (from UC_002).
	1.1	System displays: - Basic Info: Common name, scientific name, plant family - Visual Guide: High-quality images of the plant at different growth stages - Care Instructions: • Watering frequency (e.g., "2x/week in dry season") • Sunlight needs (e.g., "Bright indirect light") • Soil type (e.g., "Well-draining mix") - Local Adaptations: • Monsoon season adjustments • Air-conditioned environment tips
	1.2	Registered users can: - Save plant to "My Garden" - Set up care reminders (links to UC2.1)
	Post-Condition	User accesses all critical care information; optional setup for reminders.

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Alternate Flow	Pre-Condition	User wants troubleshooting help (plant looks unhealthy).
	2.1	System expands "Common Issues" section: - Symptom checker (e.g., "Yellow leaves? → Overwatering") - Image-based pest/disease identification - Links to relevant forum discussions (UC3.1)
	2.2	User selects a symptom to view step-by-step recovery guide.
	Post-Condition	Targeted troubleshooting advice is displayed.
Robust Flow	Pre-Condition	Database fails to load full plant details.
	3.1	System shows: - Cached basic info (name, care level) - Message: "Detailed data unavailable. Try again later." - Option to browse similar plants
	3.2	Automatically retries fetching data in background.
	Post-Condition	User receives partial information with recovery option.

Table 11: Use Case Statement for View Plant Details

Test Case for UC_003: View Plant Details

Tested By	Mohamad Farris Foo	Test level	System Testing
Test Type	Black Box Testing – Functional & AI Validation		
Test Design Technique	Use Case Testing		

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Risk Number	2 (Medium)	
Test Case Number	TC_003.1	
Test Case Name	Plant Details Display & Troubleshooting Test	
Requirement Traceability	REQ_F301, REQ_F302	
Test Case Description	To verify the system correctly displays plant care information and troubleshooting guides for Malaysian urban plants.	
Item(s) to be tested		
1	Plant Identification Module	
Pre-condition	Plant successfully identified via UC_002	
Test Procedure Specification		
Input	Test Data	Expected Output/Result
Monstera deliciosa ID	Pre-identified plant	Displays: - "Monstera deliciosa (Lidah Mertua)" - 3 growth stage images - "Water: 1x/week in KL condos" - "Soil: Well-draining mix with perlite"
Location: Penang (high humidity)	Snake plant profile	Shows: - "Reduce watering to 2x/month during NE monsoon" - "Increase ventilation to prevent fungus"
User reports "yellow leaves"	Pothos plant profile	Displays: - Diagnostic flowchart - "Likely causes: Overwatering (70%) or nutrient deficiency (30%)" - Links to UC_007 forum threads
Simulate database timeout	Orchid profile	Shows cached basic info + "Detailed care data temporarily unavailable" Retry button visible
Post-condition	All plant data sections render completely	

Table 12: Test Case for View Plant Details

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4.1.2 Use Case: Personalized Care Management Module

Use Case ID	UC_004	
Use Case Name	Set Up Personalized Care Schedule	
Description	Generates and customizes a plant care schedule based on species, user's location, and living conditions.	
Primary Actor	Registered User	
Include use cases	UC_003 (View Plant Details)	
Scenario	Step	Action
Main Flow	Pre-Condition	User has a plant saved in "My Garden" (from UC_003).
	1.1	User selects a plant from "My Garden" and taps "Create Care Schedule."
	1.2	System auto-generates a baseline schedule using: - Plant species needs - Malaysia's climate data - User's location (GPS or manual input)
	1.3	User customizes: - Location Type (Indoor/Outdoor/Balcony) - Light Conditions (e.g., "My balcony gets 2 hours of direct sun") - Water Source (e.g., tap water/rainwater)
	1.4	System adjusts recommendations (e.g., reduces watering frequency for humid areas).
	1.5	User confirms schedule; reminders are set up.
	Post-Condition	Personalized care plans are active; reminders are scheduled.

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Alternate Flow	Pre-Condition	User disagrees with system recommendations.
	2.1	User manually edits: - Watering frequency - Fertilizer dates - Pruning reminders
	2.2	System logs adjustments to improve future suggestions.
	Post-Condition	Custom schedule is saved.
Robust Flow	Pre-Condition	No internet connection during setup.
	3.1	System uses last-known climate data and generic tropical plant defaults.
	3.2	Notifies: "Schedule created offline. Sync later for weather updates."
	3.3	Automatically adjusts reminders once online.
	Post-Condition	Basic schedule is active; updates pending reconnection.

Table 12: Use Case Statement for Set Up Personalized Care Schedule

Test Case for UC_004: Personalized Care Management Module

Tested By	Amirul Ehsan	Test level	System Testing
Test Type	Black Box Testing – Functional & AI Validation		
Test Design Technique	Use Case Testing		
Risk Number	3 (High)		

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Test Case Number	TC_004.1	
Test Case Name	Care Schedule Generation & Customization	
Requirement Traceability	REQ_F401, REQ_F402	
Test Case Description	To verify the system generates and adapts plant care schedules based on Malaysian urban conditions.	
Item(s) to be tested		
1	Plant Identification Module	
Pre-condition	Plant successfully identified via UC_002	
Test Procedure Specification		
Input	Test Data	Expected Output/Result
Monstera deliciosa ID	Pre-identified plant	Displays: - "Monstera deliciosa (Lidah Mertua)" - 3 growth stage images - "Water: 1x/week in KL condos" - "Soil: Well-draining mix with perlite"
Location: Penang (high humidity)	Snake plant profile	Shows: - "Reduce watering to 2x/month during NE monsoon" - "Increase ventilation to prevent fungus"
User reports "yellow leaves"	Pothos plant profile	Displays: - Diagnostic flowchart - "Likely causes: Overwatering (70%) or nutrient deficiency (30%)" - Links to UC_007 forum threads
Simulate database timeout	Orchid profile	Shows cached basic info + "Detailed care data temporarily unavailable" Retry button visible
Post-condition	All plant data sections render completely	

Table 13: Test Case Statement for Care Schedule Generation & Customization

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Use Case ID	UC_005	
Use Case Name	Receive Adaptive Reminders	
Description	Delivers context-aware plant care alerts that adjust based on real-time environmental data and user behavior.	
Primary Actor	Registered User	
Include use cases	UC_004 (Set Up Care Schedule)	
Scenario	Step	Action
Main Flow	Pre-Condition	User has active care schedules in "My Garden" (from UC_004).
	1.1	System checks for pending tasks: - Watering - Fertilizing - Pruning/Pest checks
	1.2	Cross-references with: - Local weather API (e.g., skips watering if rain forecasted) - User location (indoor/outdoor status) - Plant dormancy cycles (e.g., reduces winter fertilizing)
	1.3	Sends push notification with: - Action ("Water your Snake Plant today") - Context ("KL humidity: 85% → Reduce amount") - Quick actions ("Snooze 1 day" / "Mark Done")
		Logs user responses to improve future timing.
	Post-Condition	User completes task or reschedules; system updates plant health log.
	Pre-Condition	User finds reminders too frequent/infrequent.
Alternate Flow	2.1	User taps "Adjust Reminders" in notification.

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	2.2	System shows: - Current frequency vs. recommended - Slider to increase/decrease interval
	2.3	Applies changes only to this plant type.
	Post-Condition	Future reminders use new preferences; AI notes deviation from scientific guidelines.
Robust Flow	Pre-Condition	Weather API unavailable or device offline.
	3.1	Uses last cached weather data (≤ 24 h old) with disclaimer: "Using yesterday's weather".
	3.2	If no cache, falls back to: - Default tropical climate profile - User's historical adjustments
	3.3	Queues sync for when connectivity resumes.
	Post-Condition	Reminders sent with reduced accuracy; full service restores when online.

Table 14: Use Case Statement for Receive Adaptive Reminders

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Test Case for UC_005: View Plant Details

Tested By	Mohamad Farris Foo	Test level	System Testing
Test Type	Black Box Testing – Functional & AI Validation		
Test Design Technique	Use Case Testing		
Risk Number	3 (High)		
Test Case Number	TC_005.1		
Test Case Name	Reminder Delivery Accuracy Test		
Requirement Traceability	REQ_F005, REQ_F006		
Test Case Description	To verify the system delivers plant care reminders based on real-time environmental data and user behavior.		
Item(s) to be tested			
1	Reminder Notification System		
Pre-condition	User has active care schedules in Play Based™ .		
Test Procedure Specification			
Input	Test Data	Expected Output/Result	
Real-time weather data (e.g., high humidity) + User location (indoor)	Real-time data fetched. Plant cycle: Raining.	System generates reminder: "Water your Snake Plant. Rt. humidity: 88% + Reduce amount" with Quick Actions ("Snooze 1 day" / "Take Door").	
User clicks "Adjust Reminders" in notification. Current	Slider adjusted to 5 days. Plant: Snake Plant only.	UI updates: "Changes applied to Snake Plant only. New interval: Every 5 days." Future reminders reflect this change.	

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frequency: Every 3 days.		
Device offline. Last cached weather: 26°C, 80% humidity (24h old).	Default tropical climate profile active.	Notification: "Using cached data: Yesterday's weather (26°C, 80% humidity). Accuracy reduced." Reminder sent with fallback data.
Humidity=10% (arid), Temp=35°C. Plant: Cactus.	Extreme weather + no historical adjustments.	Notification: "Delay watering. Extreme dryness detected. Water sparingly." No Quick Actions (outdoor plant).
Post-condition	Offline reminders note deviation from scientific guidelines.	

Table 15: Test Case Statement for Reminder Delivery Accuracy Test

Use Case ID	UC_006	
Use Case Name	Adjust Care Tasks	
Description	Allows users to manually modify or override automated care schedules based on observed plant conditions.	
Primary Actor	Registered User	
Include use cases	UC_004 (Set Up Care Schedule), UC_005 (Receive Adaptive Reminders)	
Scenario	Step	Action
Main Flow	Pre-Condition	User has active care schedules in "My Garden" (from UC_004).
	1.1	User selects a plant and taps "Adjust Care."
	1.2	System displays: - Current automated schedule - Recent weather impacts (e.g., "Reduced watering due to monsoon") - Plant health indicators (last watered, growth progress)
	1.3	User modifies: - Watering frequency (e.g., "Every 5 days" → "Every 7 days") - Task type (add/remove fertilizing, pruning)

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		- Custom notes (e.g., "Added mulch, reduce water")
		System validates changes (e.g., prevents overwatering succulents) and confirms update.
	Post-Condition	Modified schedule triggers revised reminders (UC_005); AI logs user override for learning.
Alternate Flow	-	
Robust Flow	-	

Table 16: Use Case Statement for Adjust Care Tasks

Test Case for UC_006: View Plant Details

Tested By	Amirul Ehsan	Test level	System Testing
Test Type	Positive/Negative Scenarios, Boundary Value Analysis		
Test Design Technique	Use Case Testing		
Risk Number	3 (High)		
Test Case Number	TC_006.1		
Test Case Name	Adjust Care Task Test		
Requirement Traceability	REQ_F006, REQ_F007		
Test Case Description	Verify users can modify care schedules, and overrides are handled correctly.		
Item(s) to be tested			
1	Reminder Notification System		
Pre-condition	User has active plant (e.g., Snake Plant) with schedule (UC_004).		
Test Procedure Specification			

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Input	Test Data	Expected Output/Result
User taps "Adjust Care" for Snake Plant.	Current schedule: Water every 5 days. Weather impact: "Reduced watering due to high humidity."	Display: Current schedule, weather impact, health indicators. Editable fields enabled.
User changes watering to "Every 7 days."	Input: 7 days (valid for Snake Plant).	System saves changes. Confirmation: "Schedule updated. Next watering in 7 days." Revised reminder triggered (UC_005).
User sets watering to "Every 1 day" for Succulent.	Input: 1 day (overwatering risk).	Validation error: "Overwatering risk! Max interval: 3 days for succulents." Changes rejected.
User adds note: "Added mulch, reduce water."	Note length: 50 chars (within limit).	Note saved and appended to plant profile. AI logs override for learning.
New plant with no schedule. Weather API offline.	Default schedule: "Tropical profile."	Display: "No existing schedule. Set defaults?" Allows manual entry.
Post-condition	Modified schedules trigger revised reminders (UC_005).	

Table 17: Test Case Statement for Adjust Care Task Test

4.1.3 Use Case: Community Platform Module

Use Case ID	UC_007	
Use Case Name	Create/View Forum Post	
Description	Enables users to participate in community discussions by creating posts or viewing existing threads about plant care.	
Primary Actor	Registered User	
Include use cases	UC_003 (View Plant Details)	
Scenario	Step	Action
Main Flow	Pre-Condition	User is logged in and accesses the Community section.
	1.1	User selects "New Post" and chooses:

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		- Post type (Question/Showcase/Tip) - Related plant (auto-linked to UC_003)
	1.2	Composes content with: - Text (min. 20 characters) - Optional photos (max 3 images, 5MB each) - Tags (#Pests #BeginnerHelp etc.)
	1.3	System performs auto-moderation: - Flags offensive language - Suggests similar existing threads
	1.4	Post publishes to relevant category with timestamp.
	Post-Condition	Post appears in forum; others can respond.
Alternate Flow	Pre-Condition	User struggles to describe plant issue.
	2.1	Selects "Help Me Describe" → System provides: - Symptom checklist (leaf color/texture) - Template: "My [plant] has [issue] in [location]"
	2.2	Auto-attaches plant profile if identified from photos.
	Post-Condition	Structured post with higher chance of accurate help.
Robust Flow	Pre-Condition	Post contains blocked keywords or blurry photos.
	3.1	System holds post for review with explanation: - "Your image is unclear - try retaking in daylight" - "The term '[word]' violates guidelines"
	3.2	Allows editing before resubmission.

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	Post-Condition	Cleaned post publishes after corrections.

Table 18: Use Case Statement for Create/View Forum Post

Test Case for UC_007: Community Forum Post

Tested By	Mohamad Farris Foo	Test level	System Testing
Test Type	Positive/Negative Scenarios, Boundary Value Analysis		
Test Design Technique	Use Case Testing		
Risk Number	3 (High)		
Test Case Number	TC_007.1		
Test Case Name	Create/View Forum Post Test		
Requirement Traceability	REQ_F007, REQ_F008		
Test Case Description	Verify users can create, auto-moderate, and structure forum posts correctly.		
Item(s) to be tested			
1	Post creation UI		
2	Auto-moderation (text/image)		
3	Symptom checklist integration		
Pre-condition	User is logged in.		
Test Procedure Specification			
Input	Test Data	Expected Output/Result	
User selects "New Post" → "Question".	Post Type = "Question"	System enables text input field and image upload button	
User enters valid text (15 characters)	Text = "Why are my plant's leaves yellow?"	System accepts input (no error messages)	
User uploads 2 valid images	Image1.jpg (1MB), Image2.png (0.5MB)	System displays thumbnail previews of both images	
User enters blocked keyword	Text = "This damn plant won't grow!"	System blocks submission, shows: "Term 'damn' violates guidelines"	

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Post-condition	- Approved posts appear in the forum with timestamps. - Flagged posts are logged for admin review. - Edited posts retain version history.
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Table 19: Test Case Statement for Create/View Forum Post Test

Use Case ID	UC_008	
Use Case Name	Comment/React to Forum Posts	
Description	Allows users to engage with community posts through comments and reactions.	
Primary Actor	Registered User	
Include use cases	UC_007 (Create/View Forum Post)	
Scenario	Step	Action
Main Flow	Pre-Condition	User is viewing an existing forum post (from UC_007)
	1.1	User scrolls to the comments section.
	1.2	Chooses engagement method: - Comment: Types text (min. 10 characters)
	1.3	System validates content: - Checks for spam/offensive language - Links @mentions to user profiles
		Action is published: - Comments appear in thread - Reactions update post tally
	Post-Condition	Engagement is visible to the community; original poster receives notification.
Alternate Flow	-	
Robust Flow	-	

Table 20: Use Case Statement for Comment/React to Forum Posts

Test Case for UC_008: Community Forum Post

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Tested By	Amirul Ehsan	Test level	System Testing
Test Type	Functional, Positive/Negative Scenarios		
Test Design Technique	Use Case Testing		
Risk Number	3 (High)		
Test Case Number	TC_008.1		
Test Case Name	Comment/React to Forum Posts Test		
Requirement Traceability	REQ_F009, REQ_F010		
Test Case Description	Verify users can comment/react to forum posts with proper validation and notifications.		
Item(s) to be tested			
1	Comment submission functionality		
2	Reaction (emoji) system		
3	Input validation (text length, blocked words)		
4	@mention linking		
5	Notification system		
Pre-condition	- User is logged in. - Forum post exists (from UC_007). - Auto-moderation rules (blocked words, min length) are active.		
Test Procedure Specification			
Input	Test Data	Expected Output/Result	
User types valid comment	Text="This helped! My Monstera improved." (25 chars)	System accepts input; "Post Comment" button enables.	
User selects reaction (👍)	Reaction=Thumbs Up	Reaction count increments by 1 on post. No notification sent.	
User types short comment	Text="Thanks!" (7 chars)	Error: "Comment must be ≥10 characters." Submit disabled.	
User uses @mention	Text="@PlantExpert check this!"	@PlantExpert links to user profile (blue clickable text).	
User enters blocked word	Text="This post is damn wrong!"	Error: "Term 'damn' violates guidelines." Comment held for review.	
Post-condition	- Valid comments appear in the thread.		

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	- Original poster receives notifications for comments (not reactions).
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Table 21: Test Case Statement for Comment/React to Forum Posts Test

Use Case ID	UC_009	
Use Case Name	Report Forum Content	
Description	Allows users to flag inappropriate posts/comments for moderator review.	
Primary Actor	Registered User	
Include use cases	UC_007 (Create/View Forum Post), UC_008 (Comment/React to Posts)	
Scenario	Step	Action
Main Flow	Pre-Condition	User is logged in and accesses the Community section.
	1.1	User clicks : → "Report" on offending content.
	1.2	System displays Malaysia-specific categories: • Racially Insensitive • Dangerous Plant Advice (e.g., toxic DIY pesticides) • Fake Nursery Ads
	1.3	User selects category + adds context (optional 50 chars).
	1.4	System: • Validates against duplicate reports • Awards 2 Karma Points if first reporter • Notifies moderators via Telegram bot (priority for safety alerts)
	Post-Condition	Content is hidden from reporter; case appears in admin dashboard (UC_011) within 5min.
Alternate Flow	Pre-Condition	User attempts to report scientifically valid content (e.g., proper neem oil usage).

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	2.1	System detects conflict with plant database facts.
	2.2	Shows popup: "This advice matches our experts' guidelines. Learn why [Expand]".
	Post-Condition	Either: a) User gains "Fact-Check" badge for reconsidering, OR b) Case routes to Tier-2 moderators.
Robust Flow	Pre-Condition	User submits report during internet outage.
	3.1	Auto-retries submission every 5min for 2hrs.
	3.2	If unsent after 2hrs, notifies: "Report pending - connect to WiFi".
	Post-Condition	Report syncs when online; timestamp preserved for moderation queue.

Table 22: Use Case Statement for Report Forum Content

Test Case for UC_009: Report Forum Contents

Tested By	Mohamad Farris Foo	Test level	System Testing
Test Type	Functional, Security		
Test Design Technique	Use Case Testing		
Risk Number	5 (Critical)		
Test Case Number	TC_009.1		
Test Case Name	Report Forum Content Test		
Requirement Traceability	REQ_F011, REQ_F012		
Test Case Description	Verify users can report if there's any bad post or reaction.		
Item(s) to be tested			

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1	Report submission UI	
2	Category validation (Malaysia-specific)	
3	Duplicate report detection	
4	Offline submission handling	
5	Conflict detection with scientific guidelines	
Pre-condition	User is logged in with reporting privileges.	
Test Procedure Specification		
Input	Test Data	Expected Output/Result
User submits valid report	Category=Toxic Plant Advice, Notes="Recommends harmful pesticides"	System: - Deduplicates - Awards 2 Karma Points - Hides content from reporter - Sends Telegram alert to mods
User reports scientific content	Category=Misinformation on proper neem oil usage	System detects conflict with guidelines, shows: "This matches expert advice. Learn why [link]."
Offline report submission	Network disconnected during submission	System queues report, retries every 5min for 2hrs. After 2hrs, prompts: "Submit via WiFi?"
Duplicate report attempt	Same user reports same content twice	System rejects: "You’ve already reported this." No karma awarded.
Post-condition	- Valid reports appear in moderator dashboard within 8 minutes. - Scientific content remains visible; user may earn "Fact-Check" badge. - Offline reports sync when connectivity resumes.	

Table 23: Test Case Statement for Report Forum Content Test

4.1.4 Use Case: Nursery & Shop Locator Module

Use Case ID	UC_010
Use Case Name	Search Nearby Nurseries
Description	Helps users find physical nurseries and gardening stores near their location.
Primary Actor	Registered User

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Include use cases		
Scenario	Step	Action
Main Flow	Pre-Condition	User enables location permissions/granted manual location input.
	1.1	User taps "Find Nurseries" → System detects: • Precise location (GPS) OR • Approximate area (IP-based if denied GPS)
	1.2	Fetches nurseries within 15km radius, prioritizing: • Stores with verified licenses • User-rated 4+ stars
	1.3	Displays interactive map with: • Real-time traffic alerts • Parking availability icons • "Plant ambulance" services (emergency pest control)
	Post-Condition	User sees at least 3 nearest options; results cached for 1hr.
Alternate Flow	Pre-Condition	Fewer than 3 nurseries found in radius.
	2.1	Expands search to 30km with toggle: "Include farther options?"
	2.2	Highlights "Plant Delivery Hubs" with same-day service.
	Post-Condition	User chooses between travel or delivery.
Robust Flow	Pre-Condition	Location services unavailable (e.g., indoor malls).
	3.1	Uses last cached location + 5km buffer.

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	3.2	Shows disclaimer: "Approximate location - refine at JomParking areas".
	Post-Condition	Results marked as "Unverified Position".

Table 24: Use Case Statement for Search Nearby Nurseries

Test Case for UC_010: Search Nearby Nurseries

Tested By	Amirul Ehsan	Test level	System Testing
Test Type	Functional, Geolocation		
Test Design Technique	Use Case Testing		
Risk Number	3 (High)		
Test Case Number	TC_0010.1		
Test Case Name	Search Nearby Nurseries Test		
Requirement Traceability	REQ_F013, REQ_F014		
Test Case Description	Verify users can find nurseries with correct location-based sorting and offline handling.		
Item(s) to be tested			
1	GPS/IP-based location detection		
2	Nursery search radius (15km/30km)		
3	Offline/cached results handling		
Pre-condition	A minimum of 3 verified nurseries exist in the test database.		
Test Procedure Specification			
Input	Test Data	Expected Output/Result	
User taps "Find Nurseries	GPS enabled (accurate to 10m) in urban area (KLCC).	Displays 3+ nurseries within 15km, sorted by: 1) Verified licenses, 2) 4+ stars.	
User denies GPS	Manual input: "Bentong, Pahang" (rural).	Uses IP-based location (~5km accuracy). Shows disclaimer: "Approximate location."	

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Fewer than 3 nurseries	Indoor location (e.g., Pavilion Mall).	Uses last cached location + 5km buffer. Marks results as "Unverified Position".
GPS unavailable	Same user reports same content twice	System rejects: "You've already reported this." No karma awarded.
Post-condition	- Results cached for 1 hour. - Unverified locations clearly flagged. - Delivery options visible when nurseries are scarce.	

Table 25: Test Case Statement for Search Nearby Nurseries

5. TEST ENVIRONMENT

This chapter section outlines the required resources that are important and part of the test environment in order to execute the testing. These resources are the hardware, software, facilities, distinctive test tools and other necessary special requirement necessary.

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5.1 Test Site

5.1.1 Developer Site

Testing will be conducted at the UniKL MIIT Software Lab (Block B, Level 3) during operating hours (8:00 AM–5:00 PM, Monday Friday), with access restricted to authorized team members.

5.2 Facilities Required

To ensure effective testing and validation of the SeedOfLife mobile application, several essential facilities have been identified and prepared as follows:

5.2.1 Lab Space

A lab area of approximately 500 square feet will be designated for testing purposes. Within this space, 200 square feet will be allocated specifically for desktop workstations, enabling testers to conduct structured and uninterrupted test procedures.

5.2.2 Furniture

The lab will be equipped with appropriate tables and chairs arranged to optimize workflow. The furniture will be positioned to encourage close interaction and collaboration, with testers seated across from each other to facilitate clear communication during testing activities.

5.2.3 Power Infrastructure

To support continuous testing operations, multiple electrical plug points will be strategically installed near each workstation. This ensures that all testing devices and systems can operate without interruption due to power constraints.

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5.2.4 Communication Facilities

The test environment will be structured to enable direct face-to-face communication among testers. This setup supports rapid decision-making, realtime issue resolution, and efficient coordination during test case execution and debugging.

5.2.5 Network Connectivity

A stable and secure internet connection will be provided within the test lab. This is critical for testing features that require external connectivity, such as AI-based food scanning, API calls, real-time data logging, and cloud-based synchronization.

5.3 Required hardware and software specifications

5.3.1 Hardware Requirements

Category	Technologies
	Mohamad Farris Foo
Device	Personal Computer (PC)
Operating System	Window 10 Pro
Browser	Opera GX / Google Chrome
Installed Memory	32 GB
Processor	12 th Gen Intel(R) Core(TM) i5-12400
Processor Speed	2.5 Ghz (3.98 Ghz Max)
System Type	64-BIT Operating System
Internal Storage	1.5TB

Table 26: Hardware Requirements

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5.3.2 Software Requirements

5.3.2.1 Database

Technologies	Descriptions
 Firebase	Google's backend service for apps. It provides authentication, real-time database, storage, and analytics. No server setup needed. Fast, scalable, and easy to use.

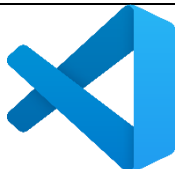
Table 27: Software Requirements - Database

5.3.2.2 Design Tools

Technologies	Descriptions
 Figma	Utilized for designing user interface (UI) wireframes and high-fidelity mockups of the mobile app.
	Used for creating visual assets such as banners, promotional materials, onboarding illustrations, and presentation slides.

Table 28: Software Requirements - Design Tools

5.3.2.3 Development Tools

Technologies	Descriptions
 FlutterFlow	Utilized for building mobile apps with Flutter through a drag-and-drop interface, FlutterFlow simplifies UI design and Firebase integration.
	Utilized for writing custom functions, debugging, and integrating AI tools via extensions (e.g., GitHub Copilot) and for seamless backend-Frontend workflows with built-in terminal, Git, and cloud/AI service support.

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Visual Studio Code	
 Android Studio	Used for Flutter and native Android app development and testing.
 Microsoft Office 365	Provides productivity tools such as Word, Excel, PowerPoint, and Outlook.

Table 29: Software Requirements – Development Tools

5.3.2.4 Programming Languages

Technologies	Descriptions
 Dart	The primary programming language used to build frontend using Flutter. It allows reactive and structured mobile UI development.
 python™	Employed for training with the AI model that powers the plant analysis feature . Used for data preprocessing, image classification, and model deployment.
 Flutter	The main cross-platform framework used to develop the mobile app for both Android and iOS.

Table 30: Software Requirements – Programming Languages

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5.4 Testing Group – Tester Involved

Test Sequence (Based on Risk Priority)	Tester	Risk Level
P5 (Critical): UC_002 Plant Identification Accuracy Test Test Sequence: 01	Mohamad Farris Foo	Critical
P5 (Critical): UC_005 Adaptive Reminders Validation Test Sequence: 02	Amirul Ehsan	Critical
P5 (Critical): UC_009 Report Forum Content Moderation Test Sequence: 03	Mohamad Farris Foo	Critical
P5 (Critical): Performance Testing - Load/Stress Scenarios Test Sequence: 04	Amirul Ehsan	Critical
P4 (High): UC_004 Personalized Care Schedule Setup Test Sequence: 05	Mohamad Farris Foo	High
P4 (High): UC_007 Forum Post Creation & Moderation Test Sequence: 06	Amirul Ehsan	High
P4 (High): API Testing - Firebase Integration Test Sequence: 07	Mohamad Farris Foo	High
P3 (Medium): UC_001 Plant Photo Upload Validation Test Sequence: 08	Amirul Ehsan	Medium
P3 (Medium): UC_006 Manual Care Task Adjustment Test Sequence: 09	Mohamad Farris Foo	Medium
P3 (Medium): UC_010 Nursery Locator (GPS/Offline) Test Sequence: 10	Amirul Ehsan	Medium
P2 (Low): UC_003 Plant Details Display Test Sequence: 11	Mohamad Farris Foo	Low
P2 (Low): UC_008 Comment/React to Forum Posts Test Sequence: 12	Amirul Ehsan	Low

Table 31: Testing Group – Tester Involved

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5.5 Preparation and training required of the test team

Roles	Responsibilities
Test Planner / Controller – Amirul Ehsan	- Ensure the SeedOfLife testing phase adheres to the schedule and meets quality standards (Section 3.7). - Define test objectives, scope, and entry/exit criteria (Sections 3.1, 3.7). - Track and report test progress during status meetings (Section 3.4). - Coordinate risk-based testing prioritization (Section 3.9.2). - Manage test cycles and resolve tester queries (Section 3.6).
Lead Tester – Mohamad Farris Foo	- Execute test cases for high-priority modules (Plant ID, Care Reminders, Payment Gateway) (Section 4.2). - Log defects and document actual vs. expected results (Section 3.5). - Validate fixes and conduct regression testing (Section 3.4). - Ensure test environment readiness (hardware/software setup) (Section 5.3). - Verify 80% test coverage before closure (Section 3.7.2).

Table 32: Preparation and training required of the test team

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6. TEST SCHEDULE

Academic Milestone	Scope of work / Activity	Date	Milestone
FYP Planning	Planning & Project Feasibility: Discuss project scope, confirm testing objectives, identify critical components (AI scanner, reminders, family module)	4 March 2025	- Feasibility Study - Early Test Planning Initiation
Proposal Defend	Requirement Study & Test Scope Finalization: Analyze all modules to extract testable features, roles, and start early documentation (SRS, use cases, test outline).	31 March 2025	- Finalize SRS - Define Testable Use Cases - Risk Analysis
Documentation Completed	Design testable UI and system architecture Begin drafting test cases for AI, dashboard, and reminders.	2 June 2025	- Initial Test Case Design - Align Test Objectives with System Design
Demonstration to Supervisor	Demonstrate early modules (login, scanner) Perform internal walkthroughs using draft test cases.	9 June 2025	- Dry Run Test - Complete Unit Test Logs
Development Progress	Continue expanding features (reminders, dashboard, admin) Execute component-level tests and prepare integration test environment.	August 2025	- Component Testing - Integration Test Planning
Testing	Execute formal test cases for all modules and user roles Record results, raise bugs, and conduct internal review of STP.	March 2025	- Acceptance Testing Phase - Finalize Test Evidence for STP

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FYP2 Presentation	Prepare complete test documentation (STP + Test Summary Report) Run final verification on Seed Of Life before demo.	June 2026	- Test Closure Report - Supervisor Sign-off - Submission & Presentation Readiness
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Table 33: Test Schedule

7. DEFINITIONS, ACRONYMS AND ABBREVIATION

7.1 Definitions

Term	Definition
Black-Box Testing	Testing without internal code knowledge; focuses on input/output.
Risk-based Testing	Prioritizing tests based on likelihood/impact of failure.
Entry/Exit Criteria	Conditions to start/stop testing (e.g., 80% test coverage).
Suspension Criteria	Conditions to pause testing(e.g., app crash, environment failure).
Acceptance Testing	Final testing to verify the system meets business requirements.
Defect Clustering	Principle that a small number of modules contain most defects.
Use Case-based Testing	Derives test causes from user workflows
V-Model	SDLC model linking development phases to testing.
Validation	Confirms the product meets user needs
Verification	Confirms software adheres to specifications.

Table 34: Definitions

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7.2 Abbreviations

Abbreviation	Full Form
SRS	Software Requirements Specification
STP	Software Test Plan
UC	Use Case
API	Application Programming Interface
GPS	Global Positioning System
UI	User Interface
ISTQB	International Software Testing Qualifications Boards

Table 35: Abbreviations

APPENDIX A – SeedOfLife Mobile Application Project Gantt Chart

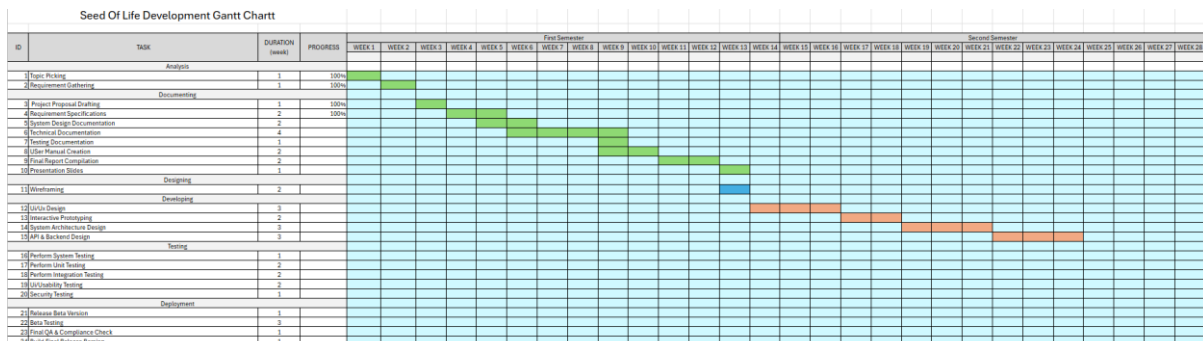


Figure 36: Appendix A- Project Gantt Chart

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